

# Desire Hernandez

Carson, CA | hernandezdesire2025@gmail.com | desirehernandez.github.io/Personal-Website/

## Technical Skills

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**Languages:** Python, SQL, JavaScript, HTML/CSS

**Frameworks:** React Native

**Developer Tools:** Git, Figma, MongoDB, JIRA, Confluence, MS Office Suite (Word, Excel, PowerPoint), Android Studio, Supabase, Node.js

**Libraries:** pandas, NumPy, Matplotlib, Scikit-learn

## Education

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**California State University, Fullerton**

Fullerton, CA

*Bachelor of Science in Computer Science, GPA: 3.47*

*Graduated: May 2025*

**Relevant Coursework:** Algorithms, Object-Oriented Programming, Data Structures, Software Engineering, Software Design, File Structure and Database, Artificial Intelligence, Software Process, Applied AI

## Experience

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**Coding Instructor**

September 2025 – Present

*Code Ninjas*

- Instruct 10+ students per session in programming fundamentals, improving problem-solving and logical thinking skills.
- Translating complex technical concepts into clear, engaging lessons while reinforcing best practices in programming.
- Collaborated with instructors and leadership to assess student skill levels, track progress, and maintain a safe, supportive learning environment.

**Data Science Research Assistant**

May 2023 - July 2023

*California State University, Fullerton | CIC-PCUBED*

*Fullerton, CA*

- Analyzed and processed datasets using Python, pandas, and NumPy to support predictive modeling research.
- Conducted research on diabetes risk prediction, ensuring data quality, consistency, and integrity.
- Contributed to model development by preparing structured datasets and validating preprocessing pipelines.

## Projects

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**TuneSwipe, AI Music Recommender** | *React Native, Spotify API, Python, Scikit-learn, Supabase, Figma* May 2025

- Collaborated in a team of four using the Scrum methodology with Jira and GitHub for sprint planning and task tracking.
- Developed a full-stack mobile app that generates personalized Spotify playlists (up to **20** songs) using real-time user swipe data and preferences.
- Designed and prototyped app UI in Figma, while documenting requirements and ensuring alignment with user needs.
- Developed and integrated swipe functionality that mapped user interactions to the backend, feeding real-time data into the recommendation engine.
- Built clustering-based recommendation model using Scikit-learn, validating accuracy through testing and iteration.

**Hospital Management System** | *UML diagrams, GRASP principles, Object-Oriented Design*

May 2024

- Applied OOAD principles to design system architecture and documented sequence diagrams and contracts.
- Modeled data flow and component interactions to ensure scalability and clarity in requirements.

**Women's Diabetes Prediction** | *Python, Scikit-learn, pandas, NumPy*

July 2023

- Built a Random Forest machine learning model to predict diabetes risk in women, achieving **96.84%** accuracy on real-world healthcare data.
- Engineered and optimized features (BMI, HbA1c, glucose, age) through data cleaning and analysis, improving model reliability and predictive performance.
- Evaluated model effectiveness using a confusion matrix, precision, and recall, delivering actionable insights for early diabetes risk detection.